Predicting Patient Insurance Costs (Regression Modeling)

Dixin Zhang

CIS-435 Practical Data Science Using Machine Learning

Professor Sunil Kakade

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Dataset: MedicalInsurance.csv (age, sex, bmi, children, smoker, region → charges)

Target variable: charges (patient insurance costs, in dollars)

1. Business Problem and Machine Learning Process

The business objective of this case study is to predict a patient's insurance costs (charges) using demographic and lifestyle indicators so that an insurer can improve pricing accuracy, and a policyholder can better anticipate expected expenses. To address this objective, the work follows a standard data mining / machine learning lifecycle:

1.1 Business understanding: Define the prediction target (insurance charges) and how prediction errors impact business decisions (pricing and budgeting).

1.2 Data understanding: Inspect the dataset schema, missingness, duplicates, and distribution of the target. The target charges is strongly right skewed, suggesting that transformations may help.

1.3 Data preparation: The process of data preparation requires three steps which include separating features and target variables, filling in missing data, transforming non-numeric data into numerical form, and adjusting numeric data to a standardized range. The target variable requires a log transformation when it is necessary to achieve both variance stabilization and skewness reduction.

1.4 Modeling: The modeling process involves building three regression models which include a baseline model, a linear model, and a non-linear model through the Scikit-Learn pipeline system.

1.5 Evaluation: The evaluation process uses R^2 and MAE and RMSE to assess model performance on both the holdout set and the cross-validation tests.

1.6 Optimization and validation strategy: The testing methods need to be modified in order to increase testing accuracy while hyperparameter optimization will be used to enhance overall system performance.

1.7 Deployment considerations: The model case chose for deployment will function as a pricing tool which requires us to track both model performance changes and fairness issues between different demographic groups.

The current approach matches standard industry practices which use predictive models to determine premium rates and execute financial assessments and forecasting (Mahaveerakannan & Arunraj, 2025).

2. Machine Learning Applications in a Related Field

2.1 Insurance cost/premium prediction and underwriting support

Businesses use predictive modeling to project their future healthcare expenses based on demographic information and lifestyle choices which include smoking habits and body mass index and age. Recent research demonstrates that ensemble models achieve better results than basic regression models when handling this specific task (Mahaveerakannan & Arunraj, 2025).

2.2 Fraud detection and anomaly detection in claims processing

Insurers can use supervised ML (e.g., gradient boosting) and unsupervised anomaly detection (e.g., Isolation Forest) to identify suspicious claims, reduce financial losses, and improve operational efficiency (Chitteti et al., 2025).

2.3 Risk stratification and proactive cost containment

Predictive analytics can classify members who are likely to become high-cost utilizers, enabling targeted case management and better resource planning. Seyam (2025) demonstrates that ML can support risk stratification with rigorous train–test and cross-validation evaluation, which is directly relevant to cost containment strategies in private insurance.

A broader industry direction is combining premium prediction with plan personalization and anomaly monitoring using AI systems (Rehman et al., 2025).

3. Machine Learning Algorithms

To provide a bit of context, case included a baseline model to serve as a reference point. In this case, a lazy Dummy Regressor just predicts the mean insurance cost across the board for all patients. The real value of more advanced models only kicks in if they can do better than that, otherwise it’s something of a moot point.

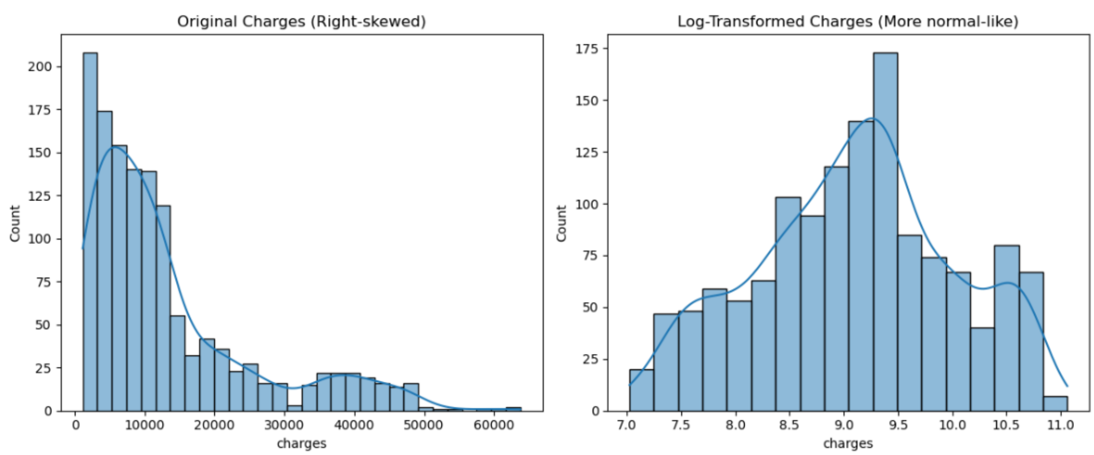
Besides, also included some Linear models like Linear Regression and Ridge Regression because they're quick to train and don't require a degree in rocket science to understand. Ridge Regression takes it a step further by adding in some regularization to keep the model stable even when it's dealing with loads of correlated predictors.

On the other hand, Non-linear ensemble models like Random Forest and Gradient Boosting are often good for structured tabular data, they can pick up on interactions and non-linear patterns without you having to think too much about them. However, the downside is that they're much harder to interpret. The age old problem of trying to balance performance and interpretability is a trade-off that's well noted in the insurance analytics literature (Mahaveerakannan & Arunraj, 2025).

4. Data Preprocessing Discussion

The data set utilized in this research is a large one with 1,338 observations. The data set has the following variables, age, sex, weight in pounds, BMI, smoke status, number of children and location. One interesting problem that we have to face is total medical costs that we have to predict. The distribution of total medical bill shows a right skewed distribution. The majority of people appear to be incuring moderate costs, while few individuals are incurring extremely high costs.

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010



Graph 4.1 Initial EDA

We implemented a Scikit-Learn pipeline for the preprocessing work because it allows us to work with multiple data types without making any bad comparisons. Starting with imputing missing values with median values that we will be standardizing into the same values as everything else. The categorical data imputation is done using the most common value for the following one-hot encoding. It is important to note that our case will be able to handle any new categories that we may have not seen before. The advantage of the execution of the pipeline is that it will prevent data leakage from occurring. (Mahaveerakannan & Arunraj, 2025)

5. Explaining Metrics

Model performance is rated using R^2, MAE and RMSE which assess different aspects of prediction accuracy. R^2 provides an estimate of how much insurance cost variations the model can explain beyond making random predictions with the average value. The method allows for model comparison but creates comprehension challenges for people who lack technical expertise. MAE provides an understanding of the actual charge difference from model predictions which is shown in dollar amounts that people can comprehend. RMSE measures errors in two ways because it assigns more significance to major errors which helps identify instances when the model fails to predict costs for high-expense patients. MAE functions as an effective daily performance assessment tool for insurance pricing models whereas RMSE helps identify rare situations that can cause significant financial losses.

	Model	R2	MAE	RMSE
7	Random Forest (log target)	0.873482	2074.178344	4431.898604
6	Gradient Boosting (log target)	0.872565	2057.149873	4447.935092
8	Extra Trees (log target)	0.853135	2167.870735	4774.989801
5	KNN (log target)	0.624177	4025.972152	7638.458513
2	Ridge (log target)	0.610058	3881.879523	7780.621059
1	Linear Regression (log target)	0.606698	3888.443159	7814.064026
0	Baseline (Dummy Mean)	-0.000919	9593.338461	12465.610442
4	ElasticNet (log target)	-0.097591	8603.157345	13053.719350
3	Lasso (log target)	-0.097591	8603.157345	13053.719350

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Best model on this holdout split: Random Forest (log target)
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Graph 5.1 Metrics

6. Interpreting Results

6.1 Holdout evaluation

The baseline model confirms that predicting the average charge is insufficient. Under an 80/20 train–test split, the baseline shows almost no explanatory power and very large errors.The baseline model (predicting the mean) performs poorly:

	Model	R2	MAE	RMSE
7	Random Forest (log target)	0.873482	2074.178344	4431.898604
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Graph 6.1 Baseline Model

This pattern is consistent with prior research suggesting that ensemble models often outperform regression-based models for insurance cost prediction due to non-linear relationships and interactions (Mahaveerakannan & Arunraj, 2025).

6.2 Changing the Testing Strategy

To demonstrate how evaluation changes with different testing strategies, the best holdout model (Random Forest) was re-tested under multiple splits. Results varied:

R^2 ranged from about 0.833 to 0.874

MAE ranged from about $2,071 to $2,367

This shows that a single train–test split can give an optimistic or pessimistic view depending on which records fall into the test set. Therefore, a more stable strategy is needed.

6.3 Cross-validation

Using 5-fold cross-validation, performance was slightly lower than the best holdout split but more stable:

	Model	R2 (mean)	R2 (std)	MAE (mean)	MAE (std)	RMSE (mean)	RMSE (std)
7	Random Forest (log target)	0.849962	0.030830	2174.272494	75.556030	4624.447142	265.787295
6	Gradient Boosting (log target)	0.848776	0.031833	2174.948279	126.780842	4641.174506	305.335699
8	Extra Trees (log target)	0.829086	0.029936	2302.223691	101.268125	4946.937230	248.232172
5	KNN (log target)	0.581621	0.030989	4128.166601	237.540443	7789.545087	376.434548
2	Ridge (log target)	0.511323	0.099820	4241.278254	258.517911	8361.904752	653.052624
1	Linear Regression (log target)	0.507514	0.100721	4251.884137	262.410783	8394.538324	661.547466
0	Baseline (Dummy Mean)	-0.002462	0.002396	9090.994473	542.603939	12085.703777	767.359808
4	ElasticNet (log target)	-0.129659	0.026563	8363.399564	452.443397	12829.667453	844.139515
3	Lasso (log target)	-0.129659	0.026563	8363.399564	452.443397	12829.667453	844.139515

--- 5-Fold CV Results (RepeatedKFold: 5 folds x 3 repeats) ---

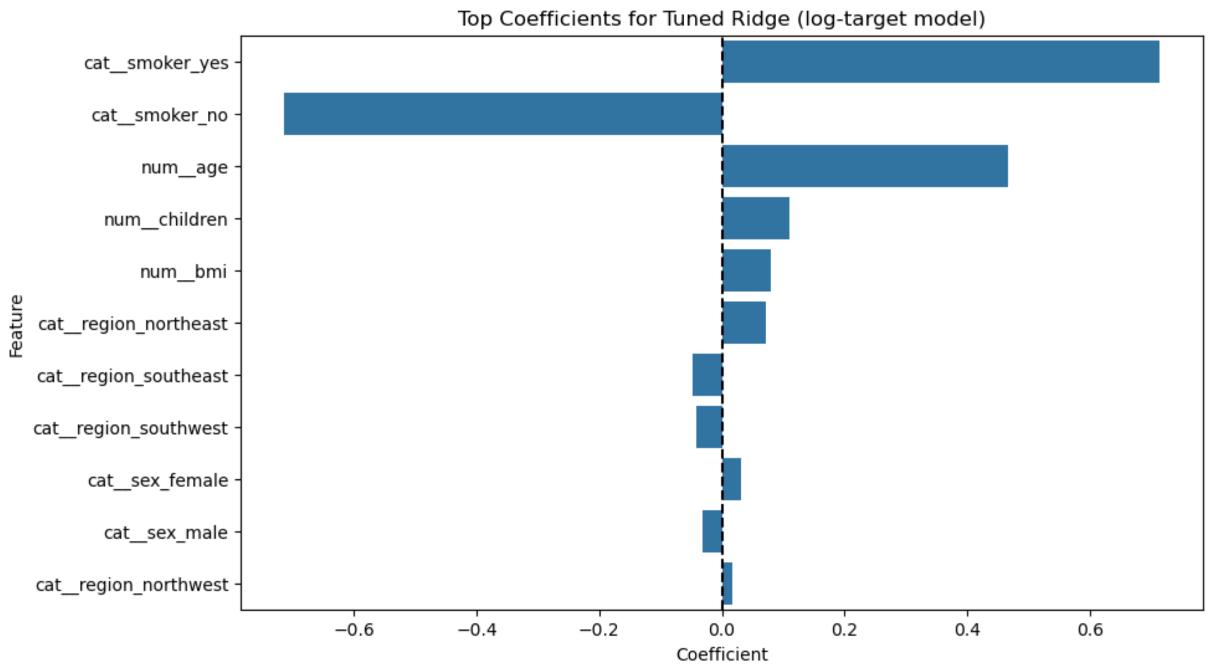
	Model	R2 (mean)	R2 (std)	MAE (mean)	MAE (std)	RMSE (mean)	RMSE (std)
7	Random Forest (log target)	0.850017	0.039211	2188.310561	232.575990	4609.139058	513.888990
6	Gradient Boosting (log target)	0.849926	0.040553	2176.529833	241.772868	4607.187918	534.349422
8	Extra Trees (log target)	0.831094	0.036229	2278.802450	211.837289	4906.968951	428.977581
5	KNN (log target)	0.574455	0.039537	4155.216810	342.019157	7857.573670	439.335858
2	Ridge (log target)	0.511597	0.091708	4246.874620	320.577730	8372.614165	612.086939
1	Linear Regression (log target)	0.507773	0.092434	4257.619340	324.053030	8405.440509	616.771354
0	Baseline (Dummy Mean)	-0.005119	0.007461	9099.272375	455.456276	12098.935800	647.578385
4	ElasticNet (log target)	-0.130712	0.040659	8366.427881	557.668290	12835.719460	807.612626
3	Lasso (log target)	-0.130862	0.040628	8368.969187	554.589919	12836.530941	806.959954

Graph 6.2 Cross-validation

Repeated 5-fold CV produced very similar mean results, but offers a better view of variability across repeated resampling. This aligns with the kind of "rigorous cross-validation procedures" recommended in related insurance ML research (Seyam, 2025).

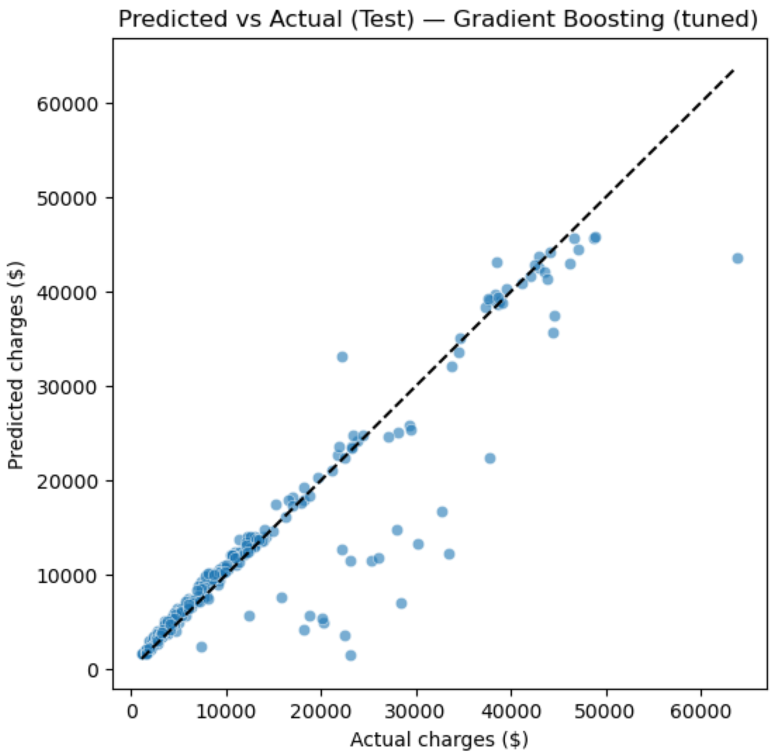
7. Recommended Steps to Optimize Model Performance

Several steps could further improve model performance and reliability. First, evaluation should prioritize cross-validation rather than a single train–test split to better capture performance variability. Second, additional hyperparameter tuning for strong ensemble models may further reduce prediction error.



Graph 7.1 Feature Engineering

Third, feature engineering could be explored to explicitly model known insurance cost relationships, such as interactions between smoking status and BMI. Finally, a real-world deployment would require ongoing performance monitoring and subgroup validation to detect data drift and reduce bias. These practices align with recent work on integrated, AI-driven insurance systems that combine prediction with monitoring and anomaly detection (Rehman et al., 2025).



Graph 7.1 Gradient Boosting

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